

Tzu-Tang Lin

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AI PLATFORM STRATEGY & SOLUTIONS ARCHITECTURE | AI R&D LEADERSHIP | AGENTIC AI & AI FOR SCIENCE

PROFESSIONAL SUMMARY

AI platform and solutions architecture professional with 8+ years of experience in AI model development, scalable research platforms, high-throughput computing, enterprise data science, and AI for Science. Currently serve as AI R&D Manager and Special Assistant to the General Manager at Repurgenesis, contributing to the company's 0-to-1 development through AI platform strategy, R&D roadmapping, cross-functional execution, and product delivery, while pursuing graduate research in Data Science at National Taiwan University's College of Electrical Engineering and Computer Science.

Previously conducted AI research at Academia Sinica, the University of Florida, and Virginia Tech, developing predictive and generative models, multimodal AI systems, biomedical AI platforms, and large-scale computational workflows. Also gained FinTech and enterprise AI experience as a Data Scientist in Cathay Financial Holdings' Data Science R&D Lab. First or co-first author of multiple peer-reviewed AI publications and first author of a 2026 NVIDIA GTC Poster Finalist submission on an AI orchestration framework.

SELECTED EXPERTISE

AI Platform Strategy, Solutions Architecture & Delivery: AI platform design, R&D roadmapping, modular architecture, use-case prioritization, cross-functional execution, POC-to-platform delivery, and executive communication

Agentic AI & Harness Engineering: AI agent and multi-agent workflow design, tool orchestration, RAG, knowledge graph reasoning, Neo4j, context and harness engineering, automated SOPs, human-in-the-loop review, LangChain, Hermes Agent, OpenClaw, and NemoClaw

AI Research & High-Throughput Computing: Predictive and generative AI, multimodal learning, biomedical AI, large-scale virtual screening, scalable research pipelines, and GPU/SLURM workflows

Technical, Local AI & NVIDIA Ecosystem: Python, PyTorch, TensorFlow, Hugging Face, GNNs, local LLM deployment and inference with Ollama, NVIDIA BioNeMo, NeMo Agent Toolkit, NIM, AWS, Docker, SQL, Linux

PROFESSIONAL EXPERIENCE

Repurgenesis Co., Ltd. — AI R&D Manager & Special Assistant to the General Manager Sep 2025 – Present

Taipei, Taiwan

AI R&D Management & Platform Development

- Lead AI platform strategy and R&D roadmapping, translating scientific workflows into prioritized platform modules, development milestones, and product plans.
- Architected the integration of four domain-specific R&D modules into a unified agentic AI platform combining scientific AI tools, RAG, knowledge graph reasoning, automated SOPs, evidence scoring, and human-in-the-loop review.
- Led the development of a scalable AI structure-based drug repurposing platform integrating proprietary datasets, automated molecular preparation, docking, affinity prediction, and candidate ranking.
- Evaluated NVIDIA BioNeMo, NeMo Agent Toolkit, and NIM for AI for Science platform architecture, agentic workflow design, and GPU-accelerated deployment; first-authored a 2026 NVIDIA GTC Poster Finalist submission ranked Top 8 of 158.

Executive Strategy & External Affairs

- Support the General Manager in the company's 0-to-1 development across corporate strategy, fundraising preparation, R&D prioritization, external partnerships, and product positioning.
- Prepare investor materials, executive presentations, speeches, and board-level AI education content, translating technical capabilities into clear strategic and business narratives.
- Coordinate corporate visits, partnership discussions, and the review and execution of MOUs and CDAs; represent the company at conferences, university events, industry meetings, and partner demonstrations.

Research Experience in the United States — Research Assistant

Aug 2023 – Aug 2025

United States

University of Florida, College of Pharmacy, AIDD Lab | May 2024 – Aug 2025

- Developed a multimodal foundation model integrating RNA sequences, molecular representations, and 3D structural information, and applied it to RNA-ligand binding affinity prediction and virtual screening.
- Built large-scale AI training, tuning, and screening workflows on a 100+ NVIDIA A100 GPU cluster, processing up to 10 million molecular inputs through automated data pipelines and SLURM-managed computing.

Virginia Tech, Department of Computer Science | Aug 2023 – Apr 2024

- Developed PathoVF, an AI bioinformatics tool integrating multimodal DNA representations, protein descriptors, protein language model embeddings, and machine learning for pathogenicity and virulence prediction.
- Built TB-scale metagenomic pipelines for sequence analysis, taxonomic profiling, and detection of antimicrobial-resistance and virulence-associated signals.

Cathay Financial Holdings Co., Ltd. — Data Scientist, Data Science R&D Lab

Dec 2022 – May 2023

Taipei, Taiwan

- Developed enterprise AI proof-of-concept solutions across FinTech and healthcare, including signature recognition, synthetic financial data generation, federated AI agents, and clinical risk prediction.

- Built an EHR-based ICU sepsis early-warning model using a Feature Tokenizer Transformer, achieving 0.915 accuracy in internal evaluation.
- Used AWS EC2, SageMaker, and S3 for scalable model development and secure data workflows; partnered with business, healthcare, and technical stakeholders and instructed Cathay General Hospital's Data Science Workshop.

Academia Sinica, Institute of Information Science — Research Assistant / Intern

Jul 2018 – Dec 2022

Taipei, Taiwan

- Led the development of AI Fleming, an AI-driven antimicrobial peptide discovery platform spanning data curation, predictive modeling, generative design, candidate prioritization, and experimental validation.
- Developed deep learning and generative models, including CNN, LSTM, Doc2Vec, ensemble learning, and WGAN-GP, for peptide activity prediction, data augmentation, and de novo sequence generation.
- Designed AI-generated peptide candidates, with 2 of 8 experimentally validated candidates demonstrating in vitro activity against cancer cells and antibiotic-resistant bacteria, with MIC values of 2–45 µg/mL.
- Integrated multiple predictive models into the public AI4AXP platform and developed the PC6 protein encoding method used in AI4AMP; contributed to work recognized with the 20th National Innovation Award.

EDUCATION

National Taiwan University, College of Electrical Engineering and Computer Science M.S. in Data Science

Expected Jun 2027

National Taiwan University

Sep 2015 – Jun 2019

B.S. in Agronomy (Specialization: Experimental Design and Biostatistics)

Undergraduate Research Fellowship, Ministry of Science and Technology, Taiwan

CERTIFICATIONS

- NVIDIA-Certified Associate: AI Infrastructure and Operations — Issued Mar 2026
- NVIDIA-Certified Associate: Generative AI Multimodal — Issued Mar 2026
- NVIDIA-Certified Associate: Generative AI LLMs — Issued Mar 2026
- NVIDIA-Certified Associate: Accelerated Data Science — Issued Mar 2026
- Neo4j Graph Data Science Certification — Issued Dec 2025

SELECTED PUBLICATIONS

1. Lin, T.-T., et al. (2023). *Intelligent de novo design of novel antimicrobial peptides against antibiotic-resistant bacteria strains*. International Journal of Molecular Sciences, 24(7), 6788. <https://doi.org/10.3390/ijms24076788>
2. Lin, T.-T., et al. (2022). *AI4AVP: An antiviral peptides predictor in deep learning approach with generative adversarial network data augmentation*. Bioinformatics Advances, 2(1), vbac080. <https://doi.org/10.1093/bioadv/vbac080>
3. Sun, Y.-Y.†, Lin, T.-T.†, et al. (2022). *Peptide-based drug predictions for cancer therapy by using deep learning*. Pharmaceuticals, 15(4), 422. <https://doi.org/10.3390/ph15040422>
4. Lin, T.-T., et al. (2021). *AI4AMP: An antimicrobial peptide predictor using physicochemical property-based encoding method and deep learning*. mSystems, 6(3), e00299-21. <https://doi.org/10.1128/mSystems.00299-21>
5. Lin, C.-Y., et al., incl. Lin, T.-T. (2026). *Harnessing sequence embedding and ensemble learning to identify antifungal peptides with low hemolytic risk*. ACS Omega. Advance online publication. <https://doi.org/10.1021/acsomega.6c00049>

Full list: Google Scholar (link above)

Patent: Lin, C.-Y., et al., incl. Lin, T.-T. (2020). Optimizing the SNP microarray probes design and construction of the analytic pipeline for SNP calling and genotyping based on the whole genome assembly. Technical know-how record No. 05T-1091014.

HONORS, LEADERSHIP & PROFESSIONAL SERVICE

- **2026 — Finalist, NVIDIA GTC 2026 Poster**, Top 8 of 158 submissions, presenting an AI-orchestrated drug repurposing platform.
- **2026 — Invited Speaker**, “Agentic Orchestration for AI-Driven Drug Repurposing and Development,” Fu Jen Catholic University Life Sciences Training Achievement Competition.
- **2023 — Co-recipient, 20th National Innovation Award, Academic Research Category**, for “AI Fleming: AI-powered platform for advanced generation, selection, and validation of functional and therapeutic peptides,” awarded by the Research Center for Biotechnology and Medicine Policy, Taiwan.
- **2023 — Instructor, Cathay General Hospital Data Science Workshop**, delivering programming and data processing training for hospital staff.
- **2023 — Bronze Medal, Kaggle Novozymes Enzyme Stability Prediction Challenge**, Top 9% worldwide.
- **2022 — Invited Speaker**, “Hunt for Novel Antibiotics: Discovering New Antimicrobial Peptides in AI,” Kiss Science event hosted by the National Science and Technology Council, Taiwan.
- **2022 — Scientific Reviewer**, *Briefings in Bioinformatics*.
- **2021–2022 — Invited Exhibitor, Healthcare+ Expo Taiwan**, showcasing AI-driven biomedical research and AI Fleming-related platform outcomes.
- **2019–2022 — Research Mentor**, supervised four summer interns at Academia Sinica, Institute of Information Science, guiding them to complete independent research projects.